

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

8875/02

Paper 2

16 September 2009

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
Total	

This document consists of **10** printed pages.



Innova Junior College

[Turn over

Section A**Answer all the questions in the spaces provided.**

- 1 (a)** Complete Table 1.1 to show two differences between the three classes of macromolecules.

Table 1.1

Macromolecules	Polysaccharides	Proteins	Nucleic Acids
Monomers			
Names of bonds linking monomers			

[2]

- (b)(i)** State two structural differences between cellulose and starch.

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..... [2]

- (ii)** Explain how these differences allow them to serve different functions in a plant cell.

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..... [2]

- (c) Although cellulose and collagen belong to different classes of biomolecules, yet they serve similar function. Describe one important similarity that these two molecules share that allow them to serve their function well.

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..... [2]

- (d) Suggest why there seems to be a greater diversity in terms of the types of proteins compared to polysaccharides.

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..... [2]

[Total: 10]

- 2 The amino acid sequences of the two proteins (enzyme **X** and **Y**) were determined. Each contained 138 amino acids joined together in a single chain. The only differences in sequence were noticed in a central part of the proteins. In Table 2.1, the amino acids concerned are given, together with their numbered individual positions in the proteins. The sequence of 'bases' (nucleotides) corresponding to this central sequence is given for protein **X**.

Table 2.1

Protein X	86	87	88	89	90	91	92	93	
N terminal	Lys	Ser	Pro	Ser	Leu	Asn	Ala	Ala	C terminal
DNA sequence	TTT	TCA	GGT	AGT	GAA	TTA	CGA	CGA	
mRNA sequence									

- (a) Enter the appropriate mRNA sequence for protein **X** in Table 2.1. [1]

- (b) Describe briefly how the DNA sequence is converted to mRNA.

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..... [3]

Further investigations showed that the enzyme **Y** was a result of **two** mutations in the DNA coding for enzyme **X**. In the first mutation, a single nucleotide was **deleted** (lost) from the DNA. In the second mutation, a single nucleotide was **inserted** (added) to the DNA.

Table 2.2

Amino Acid	mRNA codons
Lys	AAA
Ser	AGU or UCA
Pro	CCA
Leu	CUU or UUA
Asn	AAU
Ala	GCU
Val	GUC
His	CAU or CAC
Met	AUG

- (c) Using the genetic codes given in Table 2.2, state
- (i) the sequence of mRNA codons for protein Y in Table 2.3;

Table 2.3

Protein Y	86	87	88	89	90	91	92	93	
N terminal	Lys	Val	His	His	Leu	Met	Ala	Ala	C terminal
mRNA sequence									

[1]

- (ii) the **DNA base** that was deleted in the first mutation;

[1]

- (iii) the **DNA base** that was inserted in the second mutation.

[1]

- (d) In another mutation, a single nucleotide was substituted in the DNA sequence that code for protein Y. However, there was no change in the activity of the enzyme.

Explain, with the aid of relevant examples from Table 2.2, why this is the case.

[3]

[Total: 10]

- 3 In anaerobic respiration in muscle tissue the conversion of glucose to lactic acid can be represented as a four stage as in Fig. 3.1:

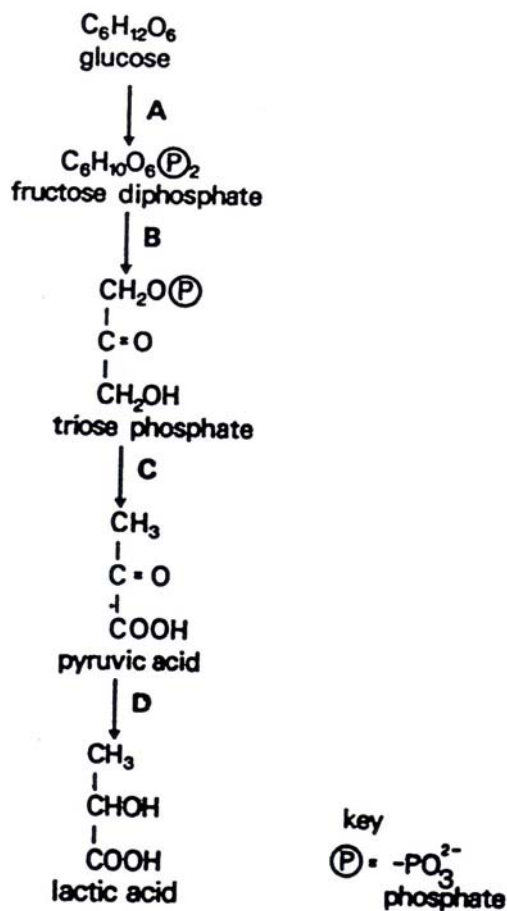


Fig. 3.1

- (a) Name the four stages in terms of the chemical processes occurring.

A

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B

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C

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D

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[2]

- (b) Is there a net change in the concentrations of NAD and NADH_2 in the whole process? Explain your answer.

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..... [3]

- (c) If one molecule of glucose is converted to two molecules of lactic acid, what is the net synthesis of ATP? Explain your answer.

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..... [3]

- (d) In what ways is the ethanol formation different from that of lactate?

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..... [2]

[Total: 10]

- 4 A length of DNA from one of a pair of homologous chromosomes is shown in Fig. 4.1. The target sites of *EcoR*I are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

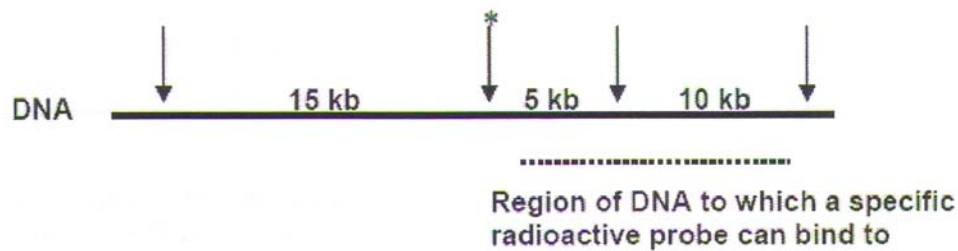


Fig. 4.1

A mutation alters one base of coding sequence at the site marked with an asterisk (*). DNA from two individuals, one who is normal and the other heterozygous for this mutation is cut with *EcoR*I and the DNA fragments separated by gel electrophoresis.

(a) On Fig 4.2 below,

- (i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individual tested and label the size of the fragments produced. [3]
- (ii) Circle the fragments which will bind to the radioactive probe and appear as black bands on an autoradiography. [1]
- (iii) Add an arrow to show the direction in which the fragments moved during electrophoresis. [1]

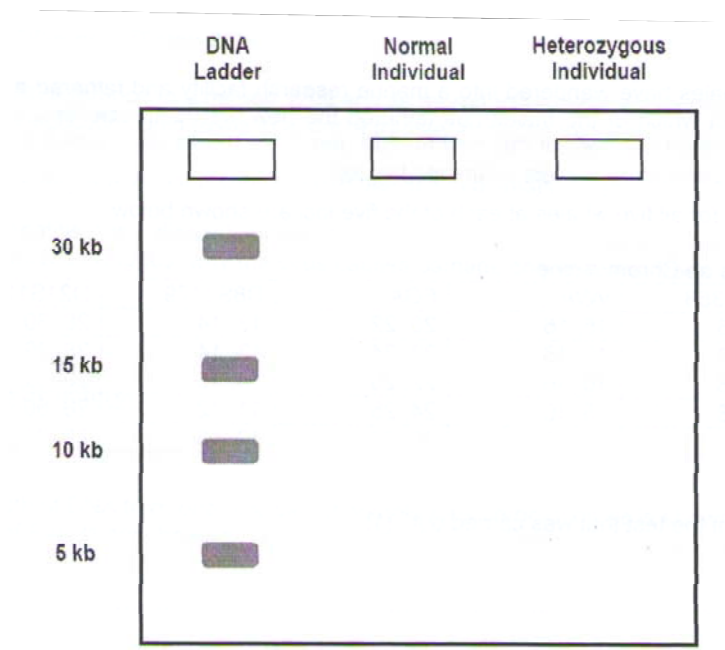


Fig. 4.2

- (b)** State the purpose of DNA ladder.

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..... [1]

- (c)** Explain how the DNA fragments can be separated by electrophoresis.

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..... [4]

[Total: 10]

Section B

Answer EITHER 5 or 6 on the Answer Paper provided.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Explain how homology (anatomical, embryological and molecular) supports Darwin's theory of natural selection. [10]

- (b)** Explain the unique features and normal functions of stem cells in a living organism, using appropriate examples to illustrate. [10]

[Total: 20]

- 6 (a)** Compare the structure and organization of prokaryotic and eukaryotic genomes. [10]

- (b)** Describe how eukaryotes regulate gene expression at chromosomal and post transcriptional level. [10]

[Total: 20]